
KnowSelf:

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness.

■ Source: arxiv.org/abs/2504.03553

The Problem: "Flood Irrigation"

Current agent training indiscriminately injects knowledge, leading to fragile planning and high costs.

- **Trajectory Imitation (e.g., ReAct)**: Blind pattern fitting
- **Trial-and-Error (e.g., Reflexion)**: Wasted steps on simple tasks
- **Knowledge-Augmented (e.g., Expel)**: Over-reliance on retrieval

The Missing Link: Situational Self-Awareness

Humans possess the metacognitive ability to assess if they can:

1. Handle a situation directly
2. Need to pause and reflect
3. Must seek external help

KnowSelf brings this capability to LLMs.

Three Situational Modes



The KnowSelf Framework

1. Data Construction

Agent self-explores. Heuristic criterion classifies situation and inserts special tokens.

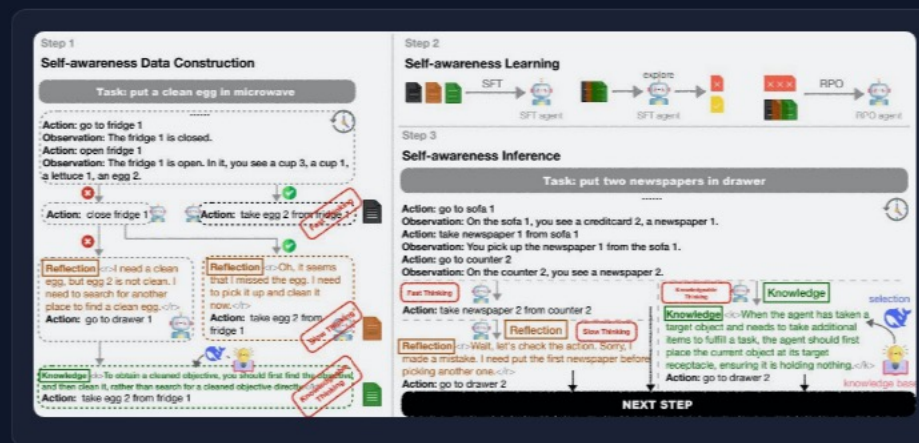
2. Two-Stage Training

SFT: Learn initial patterns.

RPO: Refine token generation policy.

3. Self-Aware Inference

Agent emits special tokens to trigger direct action, reflection, or retrieval.



KnowSelf in Action

Knowing when to reflect vs. when to seek knowledge.

"Put a hot mug in cabinet"

OpenAI-01 (Fails)

| Tries to take an egg from the microwave after opening it.

KnowSelf [Reflection Token]

Realizes it should heat the mug using the open microwave.

"Put a saltshaker in drawer"

DeepSeek-R1 (Fails)

| Reflects drawer is open, tries to close it first, gets stuck.

KnowSelf [Knowledge Token]

Retrieves rule: place object without removing unrelated items.

ALFWorld Results: Outperforming with Less



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BACKBONE	METHOD	KNOW%	PUT	CLEAN	HEAT	COOL	EXAMINE	PUT TWO	ALL
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85

Minimal Knowledge, Maximum Impact

Methods using 100% knowledge often underperform KnowSelf. Indiscriminate injection hurts efficiency.

- **Llama-8B**: 84.3% accuracy using only 15% knowledge calls.
- **Gemma-2B**: Scales down effectively, hitting 79.8% with 26% knowledge.
- **Matches GPT-4o**: Gemma-2B KnowSelf matches GPT-4o ExpelL (which uses 100% knowledge).

Accuracy vs. Knowledge Usage (Llama-8B)

